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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, GPA: 4.0

EXPERIENCE

Cruise (General Motors Autonomous Vehicles)

Warren, MI

Software Engineer Intern, Platform OS & MCU Team

Summer 2026

- Built `get_schema()`, a **Python/C API** on Cruise's PubSubMultiReader library to enforce strict IDL field types over TLV schemas, eliminating silent, high-volume type inflation (e.g., `boolean` variables inflating to `Int64`, wasting 63 of 64 bits per field) and optimizing space by dropping serialized payload footprints by up to **68.3%** across pipeline boundaries in practice.
- Root-caused a **concurrency** bug in Cruise's `infrapy C` library that had silently corrupted every frame from GOP-5 cameras for **months**, completely blocking the Embodied AI team's object-detection pipeline from ingesting camera data; designed a context-aware routing fix restoring full **data integrity** across all downstream consumers.
- Extended a **C** extension with a recursive IDL tree walker and NAL-unit-level keyframe detector over raw **H.264** Annex-B bitstreams using Exp-Golomb decoding, enabling frame-accurate bitstream seeking at scale.

Georgia Tech Research Lab (REMAP)

Atlanta, GA

Researcher, Data & Optimization Pipelines

Jan 2025 – May 2026

- Designed spatiotemporal bit-packing and compression pipelines in **C++** that reduced storage footprints by **98.5%** (**70+ GB to <1 GB**), eliminating serialization bottlenecks for downstream routing.
- Implemented **Kalman filtering** algorithms to smooth out high-variance state estimation signals, maintaining a **96.3%** reconstruction fidelity despite radical data reduction.
- Ingested **500+ GB** of noisy sequential sensor streams, isolating critical behavioral telemetry from environmental noise to reconstruct highly accurate 3D simulation inputs.

PROJECTS

Zephyr RTOS Open Source Contributions

June 2026

- Fixed a **race condition** in the **STM32** OSPI flash **driver** where the memory-mapped read path bypassed `ospi_lock_thread()` before checking peripheral state, allowing a concurrent thread to corrupt in-flight memory transfers.
- Fixed unaligned write operations in the FCB (Flash Circular Buffer) subsystem by rounding write lengths to the alignment boundary using `fcb_get_align()`, resolving test failures on **STM32** hardware with strict alignment requirements (e.g. STM32U5 16-byte writes).

RuncOS (Real-Time Mixed-Criticality OS Kernel)

Mar 2026 – June 2026

- Engineered a safety-critical **RTOS** with a dual-layer 200Hz fault monitor including volatile heartbeat detection across 4 concurrent **FreeRTOS** tasks and a 10% stack headroom threshold, triggering **deterministic** system halt within 5ms of fault detection.
- Developed a 6-task **mixed-criticality** pipeline at 1kHz with **priority-preemptive scheduling**, **zero heap allocation**, stack overflow protection, and verified output integrity via **Renode** simulation on **STM32**, with applications in guaranteeing low-latency-sensitive domains where time-critical actions must preempt low-priority work.

LDL (Low-Latency Binary Deserialization Protocol Library)

Dec 2025 – March 2026

- Designed a **deterministic** binary framing protocol in **C** over **UART** at 115,200 baud using packed structs for **zero-copy** deserialization and raw bytes cast directly into structs, minimizing latency and allocations.
- Implemented a framing and handshake protocol with 0xF0/0x0F start-code exchange for reliable stream synchronization, and a modular protocol header decoupling the data format from driver and emulator logic.
- Engineered a dual-source **POSIX** userspace **driver** with clock drift calibration and buffered timestamp sync, validated against emulated 500ppm crystal drift and ± 2 ms jitter, achieving avg 1ms sync error, with applications in high-throughput data feed parsing.

TECHNICAL SKILLS

Languages: C++20, C (Systems/Embedded), Python (C-Extensions), Rust, Bash

Systems: Concurrent Systems, RTOS, Lock-Free Synchronization, Zero-Copy Protocols, Linux Kernel/Drivers

Performance: Zero-Heap Allocation, Cache-Aware Data Structures, Cross-Compilers (GCC/Clang), Valgrind, Perf

Tooling: Bazel, Docker, POSIX, Renode (Hardware Simulation), Yocto/Bitbake

Quantitative: Probability & Statistics, Kalman Filtering, Time-Series Analysis, Graph-Based Probabilistic Models